MILITARY SPECIFICATION

Test Set, Radar Beacon AN/APM-230B

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall operationally test Receiver, Radar R-1623/APN, and Beacon Sets, Radar AN/APN-202 or AN/APN-154(V). The test set will be used as a "go/no-go" preflight and hangar, or flight-line, tester.

1.2 Classification - The Test Set, Radar Beacon AN/APM-230B shall consist of the following items:

Item	Applicable Paragraph
Test Set, Radar	3.4.1
Cable Assembly, RF	3.4.2
Antenna	3.4.3
Coupler, Antenna	3.4.4
Cabling, Antenna Coupler	3.4.5
Cover, Test Set	3.4.6

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-3432	Cable and Wire, Electrical, Power and Control, Flexible and Extra Flexible, 300 and 600 Volts.

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MIL-E-17555	Electronic, Electrical and Electro-Mechanical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-T-23336	Technical Directives, Letter Type, Preparation of
Change No. 1982 (Dept. of the Navy)	Technical Directive, Ground Support Equipment, Test Set, Radar Beacon AN/APM-230A.
NAVWEPS 16-30APM230-1	Operation and Maintenance Instruction,Test Set, Radar Beacon AN/APM-230

STANDARDS

Federal

FED-STD-595	Color

DRAWINGS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS3057	Clamp, Cable, Electrical Connector
MS3107	Connector, Plug, Electric, Quick Disconnect
MS91236	Connector, Plug, Electrical, Series N, UG-21G/U

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UNITED TELECONTROL ELECTRONICS, INC. (07450)

70C144	Antenna
70C146	Antenna
70C147	Section, Mast, Lower
70C148	Section, Mast, Upper
70C383	Test Set, Radar
70D385	Cover, Test Set

2.2 Availability of Documents - When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Parts and materials requirements shall conform to MIL-T-21200.

b. Nonrepairable subassemblies, as outlined in MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein, shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

c. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and material shall be obtained as outlined in MIL-T-21200.

3.2 Design and Construction - The equipment shall conform to the applicable requirements of MIL-T-21200 for design, construction and workmanship, unless otherwise specified herein.

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3.2.1 Total Weight - The total weight of the equipment, including cables, shall not exceed 80 pounds.

3.2.2 Form Factor - The equipment shall have maximum dimensions of 24 inches by 17 inches by 19 inches.

3.2.3 Reliability -

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 150 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during normal use.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 6000 hours with reasonable servicing and replacement of parts.

3.2.3.3 Mean Time Between Failures - The equipment shall be capable of a minimum MTBF of 500 hours.

3.2.4 Cabling and Connections - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200.

3.2.4.1 Input Power Connection - The input power connection shall consist of a flexible cable CO-03MOE (3/18) 0340 per MIL-C-3432, 8 feet, 6 inches long, permanently attached to the test set on one end and a connector plug, quick disconnect, MS3107-20-4P with clamp, electrical connector MS3057-12A on the other end.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in MIL-T-21200.

3.2.6 Nomenclature and Nameplate - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.2.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° ±10°C) (77° ±18°V)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% RH
Input Power	114 ±1.0 VAC, 400 Hz, single phase

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3.2.8 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment except as modified herein.

3.2.9 Warm-Up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum, and shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

3.2.10 Primary Input Power Requirements - The equipment shall give specified performance from the following:

AC Power, single phase, 115V, 380-400 Hz, 150 VA.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Performance Description - The equipment must provide the necessary input interrogations and output reply measurement capabilities to perform a preflight confidence test on a Radar Beacon Set when installed in an aircraft. The Radar Beacon Set is employed in two discrete modes of operation as follows:

3.3.2.1 Beacon Mode - In the Beacon Mode, the Radar Beacon Set operates in the conventional radar transponder mode, receiving single- or double-pulse X-band interrogations, and responding with single- or double-pulse replies at an "offset" X-band frequency. When this mode of operation is to be tested, the Radar Beacon Test Set must evaluate the following beacon receiver-transmitter parameters on a go/no-go basis:

a. Transmitter power

b. Transmitter frequency (X-band)

c. Receiver frequency (X-band)

d. Transponder delay

e. Reply code spacing

f. Reply rate.

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3.3.2.2 ACLS Mode - In the ACLS mode, the X-band receiver in the beacon is disabled and only the transmitter is used. The tracking radar signals are received by the R-1623/APN K_a-band receiver. These signals are processed and used to trigger the X-band beacon transmitter. Amplitude modulation appearing on the received signals as the result of the radar lobing pattern is retransmitted by amplitude modulating the X-band transmitted reply.

The test set will evaluate parameters a. and b. of paragraph 3.3.2.1 above. In order to evaluate the R-1623/APN receiver sensitivity, the test set must supply properly timed trigger pulses to the SM-658/APM Simulator for generation of synchronous K_a-band interrogations. The X-band replies received by the test set are amplified to a usable level and supplied as a video output for evaluation of any amplitude modulation by the SM-658/APM video processor circuits. Power is also supplied for operation of the SM-658/APM Simulator.

3.3.3 Functional Requirements - The following functions shall be provided in the test set.

3.3.3.1 Reply Pulse Decoder - The test set decoder shall verify the Radar Beacon Set reply code spacing on a go/no-go basis. The double-pulse transmitter reply codes are as follows:

Code	Nominal Spacing in Microseconds
1	49
2	62
3	99
4	124
5	148

A front-panel-controlled, rotary switch for reply-code selection, and an adjacent go/no-go indicator lamp shall be incorporated. When the Beacon transmitter encoded first-to-second reply pulse spacing is within selected code requirements (nominal value ±10 percent), the decoder shall cause the indicator to illuminate (go). The decoder must reject replies (indicator off: no-go) at spacing of greater than ±15 percent from nominal.

3.3.3.2 Internal Power Supply - The test set shall incorporate a power supply for providing power to the SM-658/APM simulator. The input voltage shall be ac, single phase, 115V category "C" 400 Hz, as defined in MIL-STD-704. The output voltage shall be as defined below:

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Output voltage	28 ±2 Vdc
Output current	0.50 amp min.
Regulation	±0.5 percent
Ripple	4 mV rms
Short circuit protection	Temporary short-circuit on the output shall not result in permanent damage to the power supply.

3.3.3 Simulator Control - The equipment shall provide the following signal and power outputs for operation of the SM-658/APM simulator.

(1) Trigger - The trigger output pulse circuitry shall be triggered by the Radar Beacon Test Set sync pulse and shall be shaped and delayed to trigger the SM-658/APM Simulator. Trigger pulse characteristics are:

Pulse Width	0.4 ±0.1 usec.
Pulse delay (from sync pulse)	Variable from 1.3 to 1.8 usec minimum.
Amplitude	+3.5 volts minimum.

(2) Video - A video amplifier shall be provided to amplify the output of the test set X-band rf detector. The amplifier will be used to actuate the decoder, paragraph 3.3.3.11, and to provide a gated output representative of the first beacon reply pulse of a reply pulse pair for use by the SM-658/APM. The video gate will be actuated by the test set gate pulse such that the interrogating X-band pulse and second reply pulse of the reply pair is not present at the video output terminal. The amplifier characteristics are as follows:

Input, detected RF

Pulse width	0.2 to 0.8 usec
Amplitude	60 millivolts minimum 400 millivolts maximum

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Dynamic range of output	0 to 3.5 volts minimum into a 50 ohm load
Gain	Minimum adjustment range, 5 to 20 volts/volt.
Bandwidth	0.1 usec maximum 10 to 90 percent rise time with 2 volt output. Fall time 0.3 microsecond maximum.

3.3.3.4 Simulator Power - The power supply of the test set shall supply primary power to the SM-658/APM simulator. The power supplied shall be of the following type.

27 ±2 Vdc @ 0.35 amp.

3.3.3.5 ACLS Video Gain - Operation in the ACLS mode shall result in a gain increase in the test set receiver to compensate for the reduction in power output of the Radar Beacon Set when in the ACLS mode of operation. Circuitry shall be provided to control the test set receiver gain change to +3.0 ±1.5 dB and function only when the test set MODE switch is in the ACLS position. The gain control shall be a behind-the-front-panel adjustment, normally used during periodic calibration.

3.3.3.6 Frequency Characteristics - The test set shall operate in the "X" band of frequencies as follows:

(1) Transmitter - The transmitter shall transmit coded radar interrogations as defined below and shall be adjustable to any frequency from 8500 to 9600 MHz.

(2) Receiver - The equipment receiver shall be the crystal detector - video amplifier type to receive radar signals over the frequency range from 8800 to 9500 MHz.

3.3.3.7 Signal Characteristics - The radar signal characteristics shall be as follows:

a.	Pulse width	0.5 ±0.2 microsecond
b.	Pulse rise time	0.1 microsecond maximum
c.	Pulse fall time	0.2 microsecond maximum.

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3.3.3.8 Transmitter Characteristics -

3.3.3.8.1 Frequency Stability - After calibration of the test set and a 15 minute warm-up period, the transmitter frequency shall be within ±2 MHz of the calibration for a nominal period of one hour. The above requirement is to be met under extreme service conditions.

3.3.3.8.2 Interrogation Code Characteristics - The test set shall transmit either single- or double-pulse interrogation.

3.3.3.8.3 Pulse Characteristics - The individual pulse characteristic shall be as follows for either double- or single-pulse interrogation:

Pulse width	0.5 ±0.1 microsecond
Pulse rise time	0.1 microsecond maximum
Pulse fall time	0.2 microsecond maximum.

3.3.3.8.4 Double Pulse Interrogation - The double-pulse interrogation code spacing shall have the following characteristics:

Mode 1	1.9 ±0.05 microseconds
Mode 2	2.8 ±0.05 microseconds
Mode 3	3.7 ±0.05 microseconds
Mode 4	4.6 ±0.05 microseconds
Mode 5	5.5 ±0.05 microseconds.

3.3.3.8.5 Test Transmitter Power - The nominal peak pulse available at the test set antenna terminal will be -4 dbm. This power level shall be adjustable by a front panel control labelled "POWER SET" to within ±3 db of nominal.

3.3.3.8.6 Transmitter Detuning Protection - There shall be a 6 db attenuator and a variable attenuator (Power Set) between the transmitter pulsed oscillator and the antenna transmission line.

3.3.3.8.7 PRF Generator - The interrogation pulse repetition frequency shall be 2300 ±200 pps.

3.3.3.8.8 Interrogation Control - The interrogation MODE control shall be a seven-position rotary switch located on the front panel.

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It shall be labelled as follows:

Position	Label
First	ACLS
Second	SINGLE
Third	1
Fourth	2 DOUBLE
Fifth	3
Sixth	4
Seventh	5

3.3.3.9 Receiver Characteristics -

3.3.3.9.1 Sensitivity - The test set receiver shall be capable of giving a half-scale meter deflection when a -6 dbm signal is applied to the test set antenna terminal. The test signal shall have the pulse characteristics of paragraph 3.3.3.8.3.

3.3.3.9.2 Selectivity - The test set receiver shall operate over the frequency range of 8800 to 9500 MHz. The receiver shall be of the crystal video type.

3.3.3.9.3 Decoding Capability - The test set shall have a time discriminator with a time interval of acceptance not to exceed 0.4 microsecond as maximum, or 0.25 microsecond as a minimum. During the OPERATE function, the time shall be centered about 1.1 microseconds measured from the leading edge of the coder timing pulse. During the CALIBRATE function, the start of the time interval of acceptance shall be in coincidence with the leading edge of the coder timing pulse.

3.3.3.10 Beacon Reply Rate Indication - The beacon reply rate shall be indicated in a go/no-go manner when a negative percent difference exists between the test set inerrogation rate and the beacon reply rate.

3.3.3.11 Video Amplifier Decoder - The video amplifier decoder shall provide the circuitry to verify that the Radar Beacon Set reply code spacing is within limits and shall provide the trigger and video output for the SM-658/APM Simulator.

3.3.3.12 RF Power Measurements - The expected power levels present at the antenna terminal of the test set during normal operation are as follows:

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Test Transmitter	-4 dbm nominal
Beacon Reply	-10 dbm to -4 dbm

The test set power level will be indicated by an on-the-line "PWR SET" indication with an accuracy of ±2 db of the nominal value.

The beacon reply power level will be indicated in a "go" manner if the power falls within the -10 dbm to -4 dbm power level range.

3.3.3.13 Frequency Measurements - The test set shall be capable of measuring the frequency of the transmitted and received signals to an accuracy of ±0.1%.

3.3.4 Control Functions - Front Panel -

a. ON-OFF (Power) Switch - A two-position toggle switch shall be provided for controlling the power to the test set. In the down position, labelled "OFF," no primary power shall be applied to the test set. In the up position, labelled "ON," the test set shall be fully operative.

b. OPERATE-CALIBRATE-RADAR MONITOR - A three-position rotary switch shall be provided for performing the function of switching from calibration to normal operation.

c. FREQUENCY METER - The frequency meter shall have a direct frequency readout and a front panel adjustment.

d. KLYSTRON FREQUENCY Control - The pulsed klystron frequency tuning control shall be a front panel control.

e. KLYSTRON REFLECTOR VOLTAGE Control - The klystron reflector voltage adjustment shall be a front panel control.

f. RF PEAKING Control - A control shall be provided for peaking r-f input level to test set detector.

g. POWER SET Control - The test set transmitter power level shall be adjustable from the front panel.

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h. MONITOR SENSITIVITY Control - A control shall be provided for adjusting test set video amplifier gain for Radar Monitor operation.

i. REPLY CODE Switch - A five-position rotary switch for reply codes 1 through 5.

j. MODE Switch - A seven-position rotary mode switch with the following positions: ACLS, SINGLE, 1, 2, 3, 4, 5.

3.3.5 Antenna Characteristics - The antenna shall have a minimum gain of 9 db for both the beacon receive and transmit air path. It shall be vertically polarized and have a maximum VSWR of 1.8:1 over the frequency range of 8500-9600 MHz.

3.3.5.1 Antenna Coupler - The antenna coupler shall provide a means for coupling X-band signals between the test set and the AS-2619/APN and AS-1739A/APN-154(V) antennas. The coupling means shall be shielded to provide an attenuation to interferring X-band signals.

3.3.5.1.1 Coupling Characteristics - The antenna coupler shall provide the following coupling characteristics.

(a) When used with the AS-1739A/APN-154(V) test antenna, the coupler shall couple signals from the antenna input connector to the applicable coupler output connector with an attenuation factor of 35 ±2 db over the frequency range of 8500 to 9500 MHz. Drift resulting from exposure to extreme environmental conditions shall not exceed ±1.5 dB.

(b) When used with the AS-2619/APN test antenna, the attenuation or coupling factor from the antenna input connector to the applicable coupler output connector shall be 39 ±2 dB over a frequency range of 9310 to 9415 MHz. Drift resulting from exposure to extreme environmental conditions shall not exceed ±1.5 dB.

3.4 Detail Requirements -

3.4.1 Test Set, Radar TS-2294B/APM-230 - The test set unit shall meet the following requirements:

3.4.1.1 Function - The test set shall check the parameters of Radar Beacon AN/APN-154(V), Radar Beacon Set AN/APN-202, and the components of ACLS Receiver R-1623/APN when used in conjunction

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with either beacon. The test set shall interrogate either beacon and indicate the beacon's operating characteristics to a degree of accuracy that assures flight worthiness. The test set shall also be capable of monitoring the interrogating signal and measuring the transmitted return frequency from the radar beacon. The test set shall be capable of analyzing the beacon reply signals for the following operating characteristics:

a. Beacon transmitter minimum power by go/no-go.

b. Beacon transmitter frequency by built-in frequency meter.

c. Beacon transmitter reply rate by go/no-go.

d. Beacon reply delay time by go/no-go.

e. Beacon minimum receive sensitivity by go/no-go.

When used in the ACLS mode (paragraph 3.3.2.2), the test set shall confirm beacon operation in conjunction with the ACLS receiving equipment on the aircraft.

3.4.1.2 Form Factor and Weight - The test set unit (less cover, paragraph 3.4.4) shall have maximum dimensions of 20 inches by 13-1/2 inches by 12 inches and shall weigh no more than 35 pounds.

3.4.1.3 Contents of Unit - The test set shall include the following components and assemblies:

a. Power Supply

b. Klystron

c. Frequency Meter

d. Waveguide Detector Mount

e. Waveguide Attenuator

f. Waveguide Power Divider

g. Video Receiver

h. Encoder and Modulator

i. Indicator-Functional Readout

j. Power Cable

k. Waveguide to Type "N" Adapter.

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3.4.1.4 Controls - The test set shall have the following controls:

a. Power Switch - ON-OFF

b. Function Switch - CALIBRATE-OPERATE-RADAR MONITOR

c. FREQUENCY METER and Adjustment

d. KLYSTRON FREQUENCY Control

e. KLYSTRON REFLECTOR VOLTAGE Control

f. RF PEAKING Control

g. POWER SET Control

h. MONITOR SENSITIVITY Control

i. REPLY CODE Switch

j. MODE Switch.

3.4.1.5 Electrical Connectors - The test set shall have the following electrical connections:

Designation Symbol	Receptacle Type	Function
J1	UG-160D/U	Antenna Connect
J7	MS35183-912	Synchronizing signal to external oscilloscope for test purposes.
J8	MS39012/24-0002	Trigger Pulse for Simulator, SM-658/APM.
J9	MS39012/24-0002	Video Pulse for Simulator, SM-658/APM.
J10	MS3114E10-6S	28 Vdc power for Simulator, SM-658/APM.

3.5.1.8.2 Indicators - The test set shall have the following front panel indicators.

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Reference Designation	Receptacle Type	Function
DS102	LH76/3LC15G01	Reply code indicator
M101	LH751C15RD	Power-on indicator

3.4.2 RF Cable Assembly, W1 - The RF cable assembly shall meet the following requirements:

3.4.2.1 Function - The RF cable assembly shall connect the antenna to the test set.

3.4.2.2 Form Factor - The RF cable assembly shall be 48 ±1.0 inches long.

3.4.2.3 Weight - The RF cable assembly shall weigh no more than 2.5 pounds.

3.4.2.4 Contents of Unit - The RF cable assembly shall consist of a cable with connector on each end.

3.4.2.5 Electrical Connections - The RF cable assembly shall have connectors as follows:

Reference Designation	Plug Type	Function
P1	MS91236-21E	Antenna Connection on test set mating with J1
P2	MS91236-21E	Connects to J2 on antenna

3.4.3 Antenna (70C144) - The antenna shall meet the following requirements:

3.4.3.1 Function - The function of the horn antenna is one of transmitting and receiving RF energy in the frequency range of 8500 to 9600 MHz.

3.4.3.2 Form Factor and Mounting - The antenna shall include a detachable mast for mounting the antenna to the test set. The maximum dimensions shall be 47 inches by 12 inches by 8 inches.

3.4.3.3 Weight With Mounting - The antenna with mast shall weigh no more than 4 pounds.

3.4.3.4 Contents of Antenna Unit - The antenna shall contain the following:

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Antenna, United Telecontrol Electronics, Inc. Part No. 70C146

Mast Section, United Telecontrol Electronics, Inc. Part No. 70C147

Mast Section, United Telecontrol Electronics, Inc. Part No. 70C148

3.4.3.5 Electrical Connections - The antenna shall have the following connections:

Reference Designation	Receptacle Type	Function
J2	UG-58A/U (Modified)	Connection to test set

3.4.4 Antenna Coupler - The antenna coupler shall meet the following requirements:

3.4.4.1 Connectors - Connectors for the antenna coupler shall be as follows:

Designation	Type
P1	M39012/04-0002
P2	M39012/04-0002

3.4.4.2 Weight - The weight of the antenna coupler shall not exceed 2.5 pounds.

3.4.5 Antenna Coupler RF Cabling and Connections - The equipment shall include an rf cable for connecting the antenna coupler to the Radar Beacon Test Set.

3.4.5.1 Cables, RF - The rf cable shall be as follows:

Cable Designation	Length (inches)	Function
Not assigned	262 $\pm$1	Connects antenna coupler to test set.

3.4.5.2 Connectors - Connectors for the antenna coupler cable shall be as follows:

P1	Cable	P2
UG-536 B/U	RG-223/U	UG-536 B/U

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3.4.5.3 Weight - The weight of the cable shall not exceed 1.5 pounds.

3.4.5.4 Stowage Requirements - Stowage for the antenna coupler and antenna coupler cable shall be provided by the Radar Beacon Test Set Cover CY-4722A/APM-230.

3.4.6 Cover, Test Set CY4722A/APM-230 - The test set cover shall meet the following requirements.

3.4.6.1 Function - The test set cover shall be a combination case which houses the test set unit, the RF cable assembly, the antenna, the antenna coupler, the antenna coupler cable, and the power connector cable.

3.4.6.2 Form Factor and Weight - The test set cover shall have maximum dimension of 24 inches by 17 inches by 19 inches and shall weigh no more than 32 pounds.

3.4.6.3 Color - The exterior color of the test set cover shall be orange yellow, color No. 13538 of FED-STD-595.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Test - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

a. Initial Production Tests

b. Acceptance Tests.

4.3 Initial Production Tests - One of the first ten production equipments shall be subjected to Acceptance Tests (4.4), at temperature extremes specified for Class 2 equipment by Table 1 of MIL-T-21200. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

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4.3.1 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection. For equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Tests

c. Special Tests.

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of the material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product

b. Operational Test

c. Manufacturing Run-in Test.

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Tests - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

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4.4.1.3 Manufacturing Run-in Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant point) and a minimum amplitude of ±3 g.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the mounting plane. During the 6-hour period, the equipment shall be cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipment shall be selected for sampling tests by the Government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0*
Next 50	1
Next 75	1
Next 100	1
1 for each additional 200 or fraction thereof.	

* see (4.3).

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4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that the applicable specification requirements have been met.

b. Manufacturing run-in test specified in (4.4.1.3) except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests may be required to be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipment previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of those tests required to check the characteristics in question as a result of engineering or material changes or failure reports. Test procedures previously approved for sampling tests shall be used when applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

a. Determine the cause of failure

b. Determine if the failure is an isolated case or a design defect.

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c. Submit to the procuring activity, for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting initial production tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts, such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No items, parts or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defect and resubmitted for acceptance. Before resubmitting, full particulars concerning rejection and the action taken to correct the defects found in the original shall be furnished to the Government inspector.

4.9 Equipment Changes - Changes to this specification which affect weight, installation, interchangeability, and interchangeability of maintenance parts or assemblies, require approval of the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) in accordance with MIL-T-23336, giving necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - The equipment covered by this specification shall operationally test the radar beacon and ACLS receiving equipment (paragraph 3.4.1.1) installed in the aircraft.

6.2 Performance Characteristics - Nominal performance characteristics shall be determined at input voltage of 115 ±1.0 VAC, 400 ±10 Hz. Limiting performance characteristics shall be determined at the highest and lowest values of specified input voltages. These characteristics shall be used when testing the equipment for compliance with the minimum acceptable standard of performance.

6.3 Performance Objective - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. When it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of material, parts and processes other than those specified in MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting data.

6.4 As general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification.

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b. This Specification - This specification shall have precedence over all subsidiary specifications. Any deviation from this specification, or from subsidiary specifications, shall be specifically approved in writing by the procuring activity.

d. Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the contractor for assignment of nomenclature in accordance with paragraph 3.2.6. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired option offered herein and procurement documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of packaging and packing.

c. Data requirements.

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